

AI-Based Interactive System in Human-Computer Interaction Using Natural Language Processing for Disabled People

Project Proposal

Course Title: Artificial Intelligence Project (Part 1) Proposal

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1. Introduction

Artificial Intelligence (AI) has revolutionized how humans interact with machines, particularly through natural language processing (NLP) and speech recognition technologies. In the context of inclusivity and accessibility, these innovations are increasingly vital for people with physical disabilities who face challenges in traditional human-computer interaction (HCI). This project, AI-Based Interactive System for Disabled People, explores the development of an intelligent voice-controlled interface that enables users to perform computing tasks hands-free through speech commands.

Voice-based systems are an essential component of modern assistive technology, allowing users to interact with digital environments using spoken language. However, most existing systems rely on continuous internet connectivity, which limits usability in offline or low-resource environments. This project focuses on developing an offline-capable voice recognition system that uses AI and NLP to understand, interpret, and execute user commands accurately.

The research is relevant to Artificial Intelligence, Computer Science, and STEM, combining algorithmic modeling, linguistic analysis, and system engineering. It also aligns with Sustainable Development Goals (SDG 9: Industry, Innovation, and Infrastructure) and SDG 10: Reduced Inequalities by promoting inclusive technology and equal opportunities for persons with disabilities.

By integrating advanced voice recognition algorithms (PocketSphinx, VOSK, and NLP frameworks), the project aims to create a system that improves accessibility, supports independence, and empowers disabled individuals to engage more effectively in education, work, and daily digital interaction.

2. Problem Statement

Despite advancements in AI, accessibility technology still lacks efficient offline voice recognition systems designed for people with disabilities. Most commercial voice assistants (e.g., Siri, Alexa, Google Assistant) depend on internet connectivity and cloud processing, restricting use for individuals in areas with limited connectivity or privacy concerns.

Furthermore, current systems struggle to adapt to varied speech patterns, noise environments, and physical impairments.

According to the World Health Organization (2023), over 1.3 billion people experience significant disability globally, with accessibility barriers limiting digital participation. In Malaysia and other developing nations, assistive technologies are often unaffordable or not localized to users' needs. A reliable, offline voice interaction platform could dramatically improve digital inclusivity.

The core problem addressed in this study is:

How can AI-driven offline speech recognition improve human-computer interaction for disabled individuals, ensuring accessibility, adaptability, and accuracy under diverse real-world conditions?

This issue is significant because existing models either sacrifice speed for accuracy or fail under environmental noise. The research identifies a gap in offline, low-resource, multilingual-friendly AI models that can process natural language commands effectively. Solving this challenge will enable individuals with physical limitations-particularly those unable to use traditional input devices - to perform everyday digital tasks through intuitive voice-based commands.

3. Literature Review

Recent studies demonstrate that speech recognition technologies, supported by AI, have evolved from simple voice-to-text systems to context-aware, multimodal assistants. Ambewadikar & Baheti (2020) reviewed speech recognition for disabled users and highlighted the potential of automatic speech recognition (ASR) systems to enhance accessibility

Latif et al. (2021) discussed the integration of voice technologies in healthcare, identifying speech-to-command systems as a transformative approach for users with mobility impairments.

Haeb-Umbach et al. (2021) investigated far-field ASR and found that noise and distance significantly affect model accuracy. Their findings emphasize the importance of noise-resistant models such as VOSK and Kaldi. Ok et al. (2022) explored the educational benefits of speech recognition for students with learning disabilities, showing improved interaction and

confidence through adaptive AI interfaces. Additionally, Garg et al. (2020) demonstrated how automation through speech can enhance document management and reduce manual workload.

Comparative analyses between models like PocketSphinx and VOSK reveal that offline models have advanced dramatically in recent years. VOSK, for instance, achieved up to 83.9% accuracy in noisy environments, outperforming PocketSphinx's 24.2% under similar conditions

However, few studies have explored the customization of these models for physically disabled users or evaluated their effectiveness across different age groups and noise levels.

This project addresses that research gap by implementing and testing offline speech recognition in real-world conditions. It contributes to the literature by comparing model performance across environmental noise levels, age demographics, and system adaptability - bridging AI development with assistive technology and human-centered computing.

4. Research Questions and Objectives

Research Questions

1. How can AI-based NLP and offline speech recognition enhance computer accessibility for disabled individuals?
2. What model (PocketSphinx or VOSK) delivers higher accuracy and adaptability under varied noise and speech conditions?
3. How can such systems be optimized for real-time command execution without internet dependency?

Objectives

- To develop an AI-based voice interactive system that facilitates human-computer interaction for disabled users.
- To evaluate the Word Error Rate (WER) and performance of offline open-source speech recognition systems.
- To integrate NLP for natural, context-aware communication and task automation.

- To design a user-friendly interface enabling disabled individuals to perform computer functions via speech.
- To identify future enhancement opportunities for multilingual and emotion-aware extensions.

5. Methodology

The research employs a quantitative experimental methodology, integrating software engineering and AI development techniques. The system architecture is centered on voice-based command execution, utilizing Python-based frameworks and open-source libraries such as VOSK, PocketSphinx, and NLTK for speech recognition and NLP.

Development Process:

The project follows an **Agile (Scrum)** methodology, ensuring iterative improvement and user feedback incorporation. The stages include:

1. **Requirement Analysis:** Identify user needs, including physical accessibility and offline performance.
2. **Design:** Develop a modular architecture using speech-to-text, NLP processing, and action execution layers.
3. **Implementation:** Train models with a dataset of disabled users' voice samples to improve recognition accuracy.
4. **Testing:** Evaluate model performance under two noise conditions (50-57 dB and 68-79 dB) using WER metrics.
5. **Optimization:** Fine-tune parameters for different age groups and accents.
6. **Evaluation:** Compare accuracy between models and assess user usability.

Tools and Technologies:

- Programming Language: Python
- Libraries: VOSK API, PocketSphinx, Pyttsx3 (text-to-speech), NLTK, Tkinter

- Dataset: Pre-recorded voice commands (English)
- Evaluation Metrics: Word Error Rate (WER), detection accuracy, latency

Justification:

Agile development allows flexibility for feature enhancements and user-based testing. The use of open-source offline ASR models (VOSK and PocketSphinx) ensures accessibility, cost-efficiency, and adaptability for resource-limited settings, making it ideal for assistive applications.

6. Scope and Limitations

Scope:

The project focuses on designing an offline, voice-controlled AI system tailored for users with physical disabilities. It includes developing command-based interfaces for tasks such as opening applications, browsing, and typing via voice. The system will be evaluated based on speech recognition accuracy and usability under diverse environmental conditions.

Limitations:

- The study focuses solely on the English language, limiting applicability for non-English speakers.
- Hardware variations (microphone quality) may affect accuracy.
- Limited dataset of disabled users' voices may reduce generalization.
- The system operates offline and may not support cloud-based updates or complex NLP tasks.
- Ethical concerns regarding data privacy and voice storage will be carefully managed but may restrict dataset size.

7. Expected Outcomes

The anticipated outcome is a fully functional AI-based voice interaction system that allows disabled individuals to control computers and digital devices using speech alone. Expected features include:

- Real-time speech recognition with >85% accuracy in noisy environments.
- Intuitive interface for hands-free task automation.
- Personalized assistance through adaptive NLP.
- Enhanced inclusion and independence for physically disabled users.

The project is expected to contribute significantly to assistive AI technologies by demonstrating the feasibility of offline, low-resource voice recognition. It will also inform policymakers, developers, and researchers about scalable frameworks for accessible technology aligned with SDGs promoting equality and innovation.

8. Project Plan

The project will be executed over five phases following Agile methodology:

Phase	Activities	Timeline	Deliverables
Phase 1	Literature Review & Problem Definition	Week 1-3	Report draft & research gap analysis
Phase 2	System Design & Architecture	Week 4-6	Design document & data preparation
Phase 3	Development & Model Training	Week 7-10	Prototype system & trained models
Phase 4	Testing & Evaluation	Week 11-13	Performance metrics & user feedback
Phase 5	Final Report & Presentation	Week 14-15	Final documentation & demonstration

